Two-Timescale Vehicle Association and Resource Management for C-V2X Networks

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***Abstract*—Short-timescale resource management for dense cellular-based vehicle-to-everything (C-V2X) networks inevitably incur heavy signaling overhead and system latency caused by frequent resource re-allocation. To reduce it, the most effective way is to decrease the frequency of resource allocation. But this will cause great mismatch between the quality-of-service (QoS) requirements and available resources, and further will degrade the efficiency of resources, and finally decrease the QoS of vehicles. To solve the problem and achieve a win-win situation, i.e., improving the system performance and decreasing the signaling overhead, this letter devotes itself to study a two-timescale optimization scheme by considering the mobility nature of users in C-V2X networks. The proposed scheme is in a hierarchical optimization architecture, which includes 1) long-timescale vehicle association and resource management, and 2) short-timescale power control. For the long-timescale, a matching-auction-based vehicle association, spectrum assignment and power control algorithm (MA-VASAPCA) is proposed. Given the long-timescale decision, we derive the closed-form expression for the power variable that satisfies the instantaneous QoS requirements of various services. Simulation results show that the proposed scheme can significantly enhance the system sum-rate, guarantee the QoS requirements of users and reduce the system signaling overhead.**

***Index Terms*—C-V2X, two-timescale, vehicle association, spectrum assignment, power control.**

I. Introduction

THE EMERGENCE of vehicle-to-everything (V2X) communications paves the way for intelligent transportation systems and autonomous driving [1]. Recently, together with the explosive growth of vehicles and V2X services (i.e., delay-sensitive services and delay-insensitive services), cellular-based V2X (C-V2X) networks have drawn extensive attention [2]. For C-V2X networks, the vehicular users (VUEs) co-exist with the cellular users (CUEs) in the spectrum band, resulting in grievous co-channel interference, and finally decreasing the system sum-rate performance. To tackle this, there are a lot of works focusing on efficient spectrum assignment and power control strategies to enhance the sum-rate of C-V2X networks for delay-sensitive services [3] and delay-insensitive services [4]. Furthermore, the vehicles associated with the different base stations (BSs) may cause the differentiated system sum-rate. Traditional maximum power-based vehicle association strategies result in heavy load on the BS. To cope with it, many joint advanced vehicle association and resource management strategies have been carried out recently in [5], [6].

In literature [3], [4], [5], [6], works are carried out based on short-timescale. As C-V2X networks are normally dense networks having massive connectives, utilizing short-timescale optimization strategies tends to cause frequent resource re-allocation and vehicle re-association, this will incur ultra-high signaling overhead. To address this, a lot of works have been studied on high sum-rate optimization for long-timescale [7], [8]. For instance, the work in [7] designed a long-timescale minimum distance-based vehicle association (MDVA) strategy and proposed a joint uplink and downlink resource management algorithm to improve the sum-rate of delay-insensitive services. Although the works in [7], [8] have reduced the signaling overhead, they were limited to handling delay-insensitive services. In C-V2X networks, excepting for delay-insensitive services, there are quite a lot delay-sensitive services such as security warning related information broadcasting [9]. Adopting the above fixed long-timescale strategies into delay-sensitive service scenarios directly, the instantaneous quality of service (QoS) requirements of delay-sensitive services could not be satisfied. Therefore, it is indispensable to reduce the signaling overhead and leave more space to enhance the sum-rate and guarantee the instantaneous QoS requirements of various services (i.e., both delay-sensitive and delay-insensitive services) by making a trade-off between the long and short-timescale optimization.

Inspired by the aforementioned analysis, this letter proposes a novel two-timescale (i.e., long-timescale and short-timescale) vehicle association and resource management scheme (TT-VARMS) with the aim to achieve a win-win situation, i.e., maximizing the sum-rate and in the same time decreasing the signaling overhead. This scheme is composed of two parts, i.e., 1) long-timescale vehicle association and resource management, and 2) short-timescale power control. The formulated long-timescale optimization problem is proved to be non-convex, and is difficult to solve directly. As an alternative, a matching-auction-based vehicle association, spectrum assignment and power control algorithm (MA-VASAPCA) is

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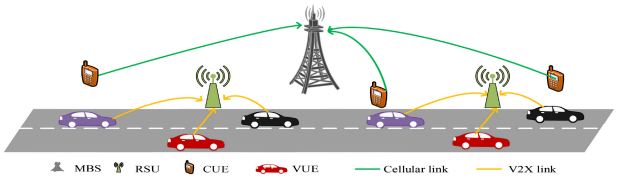


Fig. 1. System model for C-V2X networks.

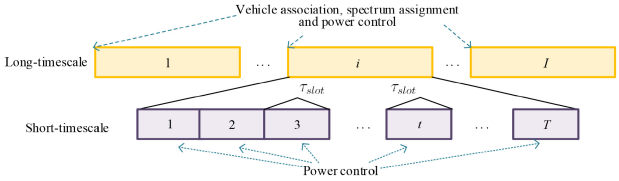


Fig. 2. The proposed two-timescale mechanism.

proposed. Given the long-timescale decision, Lagrangian dual decomposition is used to determine the closed-form expression for each vehicle's transmit power that satisfying the instantaneous QoS requirements of various services. It is worth noting that the signaling overhead of the proposed scheme lies mainly on the long-timescale. Therefore, the proposed TT-VARMS achieves a win-win situation in terms of the signaling overhead and the system sum-rate, while satisfying the instantaneous QoS requirements of various services.

II. System Model and Problem Formulation

A. System Model

This letter studies an uplink C-V2X network where the differentiated QoS requirements of both CUEs and VUEs are greatly satisfied, as shown in Fig. 1. In the network, a macro base station (MBS) is deployed to support the delay-insensitive service of CUE u, $u \in \mathcal{U} = \{1, 2, \ldots, U\}$, S road side units (RSUs) indexed by $s \in \mathcal{S} = \{1, 2, \ldots, S\}$ with coverage radius D are deployed at the roadside to support the delay-sensitive service of VUE m, $m \in \mathcal{M} = \{1, 2, \ldots, M\}$. It is assumed that there are totally F available resource blocks (RBs) and $F = U$. Hence, each CUE can utilize a single RB, i.e., no spectrum reuses between CUEs. To improve the spectrum resource utilization, U CUEs and M VUEs are allowed to share the same frequency band. Since $M \gg U$, each RB may be occupied by one CUE and multiple VUEs. Thus, the signal-to-interference-plus-noise ratio (SINR) at the MBS from the uth CUE and the sth RSU from the mth VUE can be separately expressed as

$$\gamma_u^C = \frac{P_u^{op} h_{u,MBS}}{\sum_{m \in \mathcal{M}} X_{m,u} P_m h_{m,MBS} + \sigma^2}, \quad (1)$$

$$\gamma_{m,u,s}^V = \frac{P_m h_{m,s}}{\sum_{m' \in \{\mathcal{M} \setminus m\}} X_{m',u} P_{m'} h_{m',s} + P_u^{op} h_{u,s} + \sigma^2}, \quad (2)$$

where P_u^{op} and P_m denote the transmit power of the uth CUE and the mth VUE, respectively. Specifically, we assume that P_u^{op} is fixed for each CUE and is known to all RSUs. σ^2 is the noise power, the binary variable $X_{m,u} \in \{0, 1\}$ represents the spectrum assignment indicator, if the mth VUE shares the RB with the uth CUE, $X_{m,u} = 1$, otherwise, $X_{m,u} = 0$, $h_{u,MBS}$ and $h_{m,s}$ denote the channel gain from the uth CUE to the MBS and from the mth VUE to the sth RSU, respectively, $h_{m,MBS}$ and $h_{u,s}$ are the interference channel power gain from the mth VUE to the MBS and from the uth CUE to the sth RSU, respectively.

This letter adopts a discrete time model which is divided into two-timescale, i.e., long-timescale and short-timescale as shown in Fig. 2. The long-timescale index is denoted by $i \in \mathcal{I} = \{1, 2, \ldots, I\}$, each of which includes T short scale time slots. The short-timescale index of the ith long scale time slot is represented by $t \in \mathcal{T} = \{1, 2, \ldots, T\}$, and the length of each short time slot is denoted as τ_{slot}.

B. Problem Formulation

In this letter, the objective is to maximize the sum-rate of the V2X system while guaranteeing the instantaneous QoS requirements of both CUEs and VUEs. In achieving this, a joint vehicle association, spectrum assignment and power control optimization problem is formulated as

$$\mathcal{P}1: \max_{\Phi_{m,s}^{i,t}, X_{m,u}^{i,t}, P_m^{i,t}} \sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} \sum_{s \in \mathcal{S}} \sum_{m \in \mathcal{M}} \sum_{u \in \mathcal{U}} \Phi_{m,s}^{i,t} \log_2(1 + \gamma_{m,u,s}^{V,i,t}), \quad (3a)$$

$$\text{s. t. } \log_2(1 + \gamma_u^{C,i,t}) \geq R_{th,u}^{i,t}, \forall u, i, t, \quad (3b)$$

$$\sum_{s \in \mathcal{S}} \sum_{u \in \mathcal{U}} \Phi_{m,s}^{i,t} \log_2(1 + \gamma_{m,u,s}^{V,i,t}) \geq R_{th,m}^{i,t}, \forall m, i, t, \quad (3c)$$

$$\Phi_{m,s}^{i,t}, X_{m,u}^{i,t} \in \{0, 1\}, \forall m, u, s, i, t, \quad (3d)$$

$$\sum_{s \in \mathcal{S}} \Phi_{m,s}^{i,t} = 1, \forall m, i, t, \quad (3e)$$

$$\sum_{u \in \mathcal{U}} X_{m,u}^{i,t} = 1, \forall m, i, t, \quad (3f)$$

$$0 \leq P_m^{i,t} \leq P_m^{max}, \forall m, i, t, \quad (3g)$$

in which the binary variable $\Phi_{m,s}^{i,t} \in \{0, 1\}$ is defined as the association factor at long-timescale i and small-timescale t, if the mth VUE connected to the sth RSU at the tth short scale slot of the ith long scale slot, $\Phi_{m,s}^{i,t} = 1$, otherwise, $\Phi_{m,s}^{i,t} = 0$, $R_{th,u}^{i,t}$ and $R_{th,m}^{i,t}$ represent the minimum data rate thresholds of the uth CUE and the mth VUE at the tth short scale slot of the ith long scale slot, respectively, P_m^{max} is the maximum transmit power of the mth VUE. Constraints (3b) and (3c) denote that the uth CUE and the mth VUE guarantee the minimum data rate threshold. Constraint (3d) indicates that the vehicle association and spectrum assignment decisions are binary variables. Each VUE is associated with a RSU according to constraint (3e). Constraint (3f) means that each VUE can only access to one RB that is occupied by one CUE at each short scale time slot. Constraint (3g) restricts the transmit power of the mth VUE at each scale slot. It is easy to find that $\mathcal{P}1$ is NP-hard. In addition, as C-V2X networks are normally dense networks having massive connectives, using short-timescale optimization strategy to achieve this goal will incur ultra-high signaling overhead. To solve it, we propose a low signaling overhead TT-VARMS, and the detail solution procedure is shown in the following section.

III. Proposed Two-Timescale Scheme

The proposed two-timescale scheme composes of two steps. In the first step, to reduce the signaling overhead and increase

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the system sum-rate, a joint vehicle association and resource management strategy is designed at the beginning of each long scale slot. Then, in the second step, to satisfy the instantaneous QoS requirements of users, an efficient power control strategy is proposed for each short scale slot, while the vehicle association and spectrum assignment keep unchanged.

A. Solving the Long-Timescale Problem

In this subsection, the joint vehicle association and resource optimization for long scale slots is settled down. In doing so, $\mathcal{P}1$ is transformed into the following problem

$$\mathcal{P}2: \max_{\Phi^i_{m,s}, X^i_{m,u}, P^i_m} \sum_{i\in\mathcal{I}}\sum_{s\in\mathcal{S}}\sum_{m\in\mathcal{M}}\sum_{u\in\mathcal{U}} \mathbb{E}[\Phi^i_{m,s}\log_2(1+\gamma^{V,i}_{m,u,s})], \quad (4a)$$

$$\text{s. t. } \mathbb{E}[\log_2(1+\gamma^{C,i}_u)] \geq \overline{R}^i_{th,u}, \forall u, i, \quad (4b)$$

$$\mathbb{E}[\sum_{s\in\mathcal{S}}\sum_{u\in\mathcal{U}} \Phi^i_{m,s}\log_2(1+\gamma^{V,i}_{m,u,s})] \geq \overline{R}^i_{th,m}, \forall m, i, \quad (4c)$$

$$\Phi^i_{m,s}, X^i_{m,u} \in \{0,1\}, \forall m, u, s, i, \quad (4d)$$

$$\sum_{s\in\mathcal{S}} \Phi^i_{m,s} = 1, \forall m, i, \quad (4e)$$

$$\sum_{u\in\mathcal{U}} X^i_{m,u} = 1, \forall m, i, \quad (4f)$$

$$0 \leq P^i_m \leq P^{max}_m, \forall m, i, \quad (4g)$$

where $\overline{R}^i_{th,u}$ and $\overline{R}^i_{th,m}$ represent the minimum average data rate thresholds for the uth CUE and the mth VUE at the ith long scale slot, respectively.

It can be seen that $\mathcal{P}2$ is a complicated matching problem among CUEs, VUEs, vehicle association, spectrum and power resources, which is proved to be NP-hard. To find a tractable solution, as shown in Algorithm 1, we design a MA-VASAPCA at long scale slots. Notably, the foregoing auction theory refers to the English-auction based on the rising cost rule, which consists of the following elements: 1) *Auctioneer:* S RSUs, 2) *Auction Goods*: M VUEs, 3) *Bidders*: U CUEs, and 4) *Utility Function*: $G(\Phi^i_{m,s}, u, P^i_m(\tau))$ where τ is the index of the rising cost iteration. More specifically, $G(\Phi^i_{m,s}, u, P^i_m(\tau))$ represents the satisfaction of the uth CUE sharing the same RB with the mth VUE which associated with the sth RSU, where the transmit power of the mth VUE is $P^i_m(\tau)$. $G(\Phi^i_{m,s}, u, P^i_m(\tau))$ is expressed as

$$G(\Phi^i_{m,s}, u, P^i_m(\tau)) = \Omega(\Phi^i_{m,s}, u, P^i_m(\tau)) - P^i_m(\tau), \quad (5)$$

where

$$\Omega(\Phi^i_{m,s}, u, P^i_m(\tau)) = \begin{cases} \mathbb{E}[\log_2(1+\gamma^{V,i}_{m,u,s})], \text{if (4b)} - \text{(4c)} \\ \text{are satisfied.} \\ 0, \text{otherwise.} \end{cases} \quad (6)$$

B. Solving the Short-Timescale Problem

Through the optimization scheme in Section III-A, it guarantees the average QoS requirements of various users. However, the instantaneous QoS restricts of delay-sensitive services may not be satisfied, which greatly reduces the system sum-rate and causes the extra signaling overhead. To tackle this problem, a power control algorithm is proposed to maximize the sum-rate of VUEs at each short scale slot. Hence, the optimal problem at each short scale slot is formulated as

$$\mathcal{P}3: \max_{P^{i,t}_m} f(P^{i,t}_m), \quad (7a)$$

$$\text{s. t. } \text{(3b)}, \text{(3c)}, \text{(3g)}, \quad (7b)$$

where $f(P^{i,t}_m) = \sum_{s\in\mathcal{D}^i_m}\sum_{m\in\mathcal{Q}^i_u}\sum_{u\in\mathcal{U}}\log_2(1+\gamma^{V,i,t}_{m,u,s})$. Analysis shows that the objective function $f(P^{i,t}_m)$ in (7) is a non-convex. By converting $P^{i,t}_m$ into $e^{\breve{P}^{i,t}_m}$, $f(P^{i,t}_m)$ can be equivalently transformed into a convex function which is

$$f(\breve{P}^{i,t}_m) = \frac{1}{\ln(2)}\sum_{s\in\mathcal{D}^i_m}\sum_{m\in\mathcal{Q}^i_u}\sum_{u\in\mathcal{U}}\Big(-\ln\delta + \breve{P}^{i,t}_m + \ln(h^{i,t}_{m,s})\Big), \quad (8)$$

where $\delta = \sum_{m\prime\in\mathcal{Q}^i_u, m\prime\neq m} e^{(\breve{P}^{i,t}_{m\prime} + \ln h^{i,t}_{m,MBS})} + P^{op}_u h^{i,t}_{m,MBS} + \sigma^2$.

It can be proved that (7b) and (8) are convex function and the transformed $\mathcal{P}3$ is a convex problem, which can be solved by exploiting the Lagrange dual decomposition. We define $\lambda_u \in \mathbf{\Lambda} = \{\lambda_1, \lambda_2, \ldots, \lambda_U\}$, $\mu_m \in \mathbf{u} = \{\mu_1, \mu_2, \ldots, \mu_M\}$ and $\omega_m \in \mathbf{w} = \{\omega_1, \omega_2, \ldots, \omega_M\}$ as the Lagrange multiplier variables which are associated with the constraints (3b), (3c) and (3g), respectively. By

Algorithm 1 Matching-Auction-Based Vehicle Association, Spectrum Assignment and Power Control Algorithm (MA-VASAPCA)

1: Initialize the set of the RSUs associated with the mth VUE $\mathcal{D}^i_m = \emptyset, \forall i \in \mathcal{I}$, and the set of the VUEs assigned to the uth CUE $\mathcal{Q}^i_u = \emptyset, \forall i \in \mathcal{I}$. Set the index of rising cost iteration τ, and the power increment of each iteration Δp. Initialize the transmit power of each VUE P_0.
2: *Submit bids:*
3: **for** $m = 1 : M$ **do**
4: $\tau = 0, P^i_m(\tau) = P_0$.
5: **for** $u = 1 : U$ **do**
6: **for** $s = 1 : S$ **do**
7: Calculate $G(\Phi^i_{m,s}, u, P^i_m(\tau))$ according to (5)-(6).
8: **end for**
9: **end for**
10: Place $s^* = \arg\max_{s\in\mathcal{S}} G(\Phi^i_{m,s}, u, P^i_m(\tau))$ into the set $\mathcal{D}^i_m$.
11: Sort all CUEs via the descending order of the utility function.
12: **end for**
13: *Rise cost:*
14: **for** $m = 1 : M$ **do**
15: Calculate the number of CUEs that are of most interested in the mth VUE, denoted by ρ_m.
16: **while** $\rho_m > 1$ **do**
17: $P^i_m(\tau+1) = P^i_m(\tau) + \Delta p$, $\tau = \tau + 1$.
18: Update the utility function, $P^i_m(\tau))$ and ρ_m.
19: **end while**
20: **if** $\rho_m = 1$ **then**
21: Update $\mathcal{Q}^i_u$.
22: **end if**
23: **end for**

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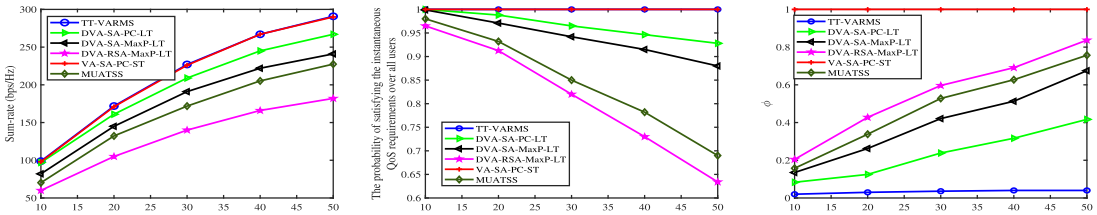


Fig. 3. The system performance (the system sum-rate, the probability of satisfying the instantaneous QoS requirements over all users, the user re-association and resource re-allocation ratio) for different number of VUEs.

Algorithm 2 Two-Timescale Vehicle Association and Resource Management Scheme (TT-VARMS)

1: **for** $i = 1 : I$ **do**
2: Do vehicle association, spectrum assignment and power control based on **Algorithm 1**.
3: **for** $t = 1 : T$ **do**
4: Compute P_m^{i,t^*} via (9)-(12).
5: **end for**
6: **end for**

utilizing Karush-Kuhn-Tucker (KKT) conditions, and taking the derivative of $\mathcal{L}(\breve{P}_m^{i,t}, \mathbf{\Lambda}, \mathbf{u}, \mathbf{w})$ with respect to $\breve{P}_m^{i,t}$, the optimal solution of (7) can be obtained. Translating $e^{\breve{P}_m^{i,t}}$ into $P_m^{i,t}$, the optimal solution is given by

$$P_m^{i,t^*} = \max\left\{\frac{1+\mu_m}{\sum\limits_{s\in\mathcal{D}_l^i}\sum\limits_{l\in\mathcal{Q}_u^i, l\neq m}\sum\limits_{u\in\mathcal{U}} \mathcal{Y} + (\lambda_u h_{u,s} + \omega_m)\ln(2)}, 0\right\} \quad (9)$$

in which $\mathcal{Y} = \frac{h_{s,l}^{i,t}(1+u_m)}{\sum_{s\in\mathcal{D}_{m\prime}^i}\sum_{m\prime\in\mathcal{Q}_u^i, m\prime\neq m} P_{m\prime}^{i,t} h_{l,s}^{i,t} + P_u^{op} h_{m,MBS}^{i,t} + \sigma^2}$.

Utilizing the subgradient method, the Lagrange multipliers in the ℓ-th iteration are updated by

$$\lambda_u^{(\ell+1)} = \lambda_u^{(\ell)} - \alpha\left(\frac{P_u^{op} h_{u,MBS}^{i,t}}{R_{th,u}^{i,t}} - \sigma^2 - \sum_{m\in\mathcal{M}} P_m^{i,t} h_{u,s}^{i,t}\right), \quad (10)$$

$$\mu_m^{(\ell+1)} = \mu_m^{(\ell)} + \beta\left(R_{th,m}^{i,t} - \log_2(1+\gamma_{m,u,s}^{V,i,t})\right), \quad (11)$$

$$\omega_m^{(\ell+1)} = \omega_m^{(\ell)} + \eta\left(P_m^{i,t} - P_m^{max}\right), \quad (12)$$

where α, β, and η are the step sizes.

C. Solution to the Original Problem and Complexity Analysis

To solve $\mathcal{P}1$, TT-VARMS is proposed as shown in Algorithm 2, which composes of solving the above two subproblems. In Algorithm 2, Algorithm 1 is executed at the beginning of the *i*th long scale slot (the length is *T*). Then, the transmit power for each short scale slot *t* in the current long scale slot *i* is adjusted to keep the signaling overhead unchanged. The transmit power control action is repeated until the power is all adjusted over all the short scale slots. For MA-VASAPCA, the computational complexity is $\mathcal{O}(M^2 + M^2\log(M) + \sum_{m\in\mathcal{M}}\rho_m)$. For the power control phase at each short scale slot, the computational complexity can be approximated by $\mathcal{O}(((U+M)U)^2)$. Therefore, the total computational complexity of the proposed TT-VARMS is $\mathcal{O}(M^2 + M^2\log(M) + \sum_{m\in\mathcal{M}}\rho_m + ((U+M)U)^2)$.

IV. Simulation Results and Analysis

Intensive simulations are done to show the performance of the proposed algorithm. All the simulations are done by using MATLAB R2020a on a computer with an Intel Core i5-8250U (1.6GHz/L3 6M) and 8GB of RAM. It is assumed that all the users are located within a target rectangular area 1000 m×1000 m. The simulation parameters are set according to 3GPP TR 36.885, where a MBS is located at the center of the considered area and a number of RSUs with $D = 150$ m are located at the roadside [10]. The number of lanes is 6, and the width of each lane is 4 m [6]. The moving speed of vehicles is set as [10, 60] km/h. The average inter-vehicle distance in the same lane is $2.5v$ m with v representing the moving speed of vehicles in meter per second [5]. Besides, we set $P_m^{max} = 17$ dBm, $P_u^{op} = 25$ dBm, $\sigma^2 = -104$ dBm. The SINR thresholds for the V2X link receiver and cellular link receiver are set to 5 dB and 0 dB, respectively. The shadowing standard deviations for the V2X link and the cellular link are 8 dB and 4 dB, respectively. The whole simulation time is $\tau_{slot} IT$ where $\tau_{slot} = 10$ ms, $I = 10$ and $T = [60, 140]$ [4], [8].

The impact of the number of VUEs (i.e., *M*) and the length of long-timescale (i.e., *T*) on the system performance (i.e., the system sum-rate, the probability of satisfying QoS requirements over all users, the system signaling overhead) are simulated, respectively. The obtained results are shown in Figs. 3 and 4. Considering that reducing the frequency of user re-association and resource re-allocation is an effective way to decrease system signaling overhead, we evaluate the system signaling overhead via the re-association and re-allocation ratio, which is $\phi = N/IT$ where *N* denotes the times of user re-association and resource re-allocation. To show the superiority of the proposed TT-VARMS, five baselines are simulated and compared: 1) DVA-SA-PC-LT algorithm, which is done in long-timescale and composed of the MDVA and our proposed resource (spectrum and power) allocation. 2) DVA-SA-MaxP-LT algorithm, which is done in long-timescale and composed of MDVA and the auction-based resource allocation algorithm in [5]. 3) DVA-RSA-MaxP-LT algorithm, which is done in long-timescale and is composed of the MDVA and random spectrum assignment algorithm with fixed maximum transmit power. 4) VA-SA-PC-ST algorithm, which is done in short timescale with the proposed MA-VASAPCA. 5) MUATSS algorithm, which is done in two-timescale adopting the match game-based user association and transmission time selection algorithm [8].

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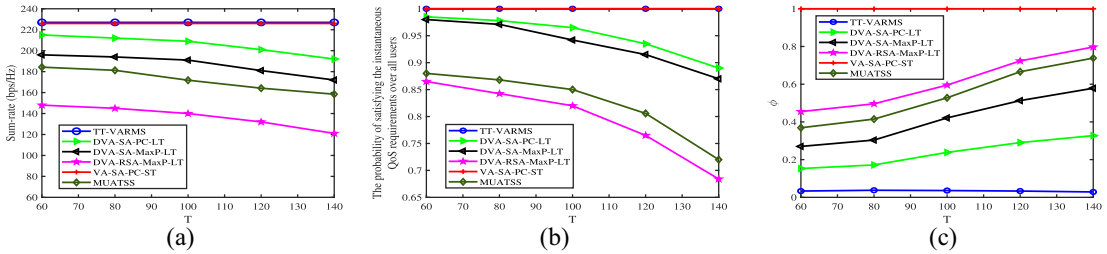


Fig. 4. The system performance (the system sum-rate, the probability of satisfying the instantaneous QoS requirements over all users, the user re-association and resource re-allocation ratio) for different length of long-timescale.

The impact of the number of VUEs on the sum-rate performance is shown in Fig. 3 (a), from which we see that the sum-rate increases as the number of VUEs increases. This is principally because that compared to the sum-rate gain brought by the increasing number of VUEs, the loss of sum-rate caused by co-channel interference among users is trivial. In addition, the proposed algorithm has a higher sum-rate than all the baselines, and the gain is around 7%-62%. Meanwhile, the proposed algorithm is capable to achieve a similar sum-rate with VA-SA-PC-ST. From Fig. 3 (b), it shows that the instantaneous QoS requirements of all the users can be satisfied by the proposed scheme and SA-PC-VP-ST, while the baselines can not guarantee the users' instantaneous QoS constraints. From Fig. 3 (c), compared to other algorithms, it is seen that among all the simulated schemes, the proposed two-timescale scheme has the lowest signaling overhead, while VA-SA-PC-ST has the highest signaling overhead. This is because that VA-SA-PC-ST guarantees the QoS requirements and improves the sum-rate based on the real-time scheduling, which leads to the enormous signaling overhead. When the number of VUEs increases, it is difficult to meet the instantaneous QoS requirements of increasing number of VUEs relying only on those long-timescale optimization algorithms. To satisfy it, user re-associated and resource re-allocated will be done, which in turn greatly increases the signaling overhead.

The impact of the length of long-timescale on the sum-rate, the probability of satisfying QoS requirements and the signaling overhead is investigated, respectively. The obtained results are included in Fig. 4. When the value of T is enlarged, adopting DVA-SA-PC-LT, DVA-SA-MaxP-LT, DVA-RSA-MaxP-LT and MUATSS, the sum-rate and the probability that the QoS requirements are satisfied decrease while the signaling overhead increases. This is because that when enlarging T, it is hard to satisfy the constraints of instantaneous QoS requirements, which reduces the sum-rate performance of the C-V2X network and accordingly the resource should be re-allocated. This finally will increase the signaling overhead. From the results, it is also seen that the proposed algorithm has the highest sum-rate and the lowest signaling overhead. This convincingly confirms the superiority of proposed two-timescale scheme.

V. Conclusion

This letter focuses investigation on the low signaling overhead oriented sum-rate maximization algorithm for C-V2X networks. Taking the diverse QoS requirements of both CUEs and VUEs into account, we formulated a sum-rate maximization problem by jointly optimizing the vehicle association, the spectrum assignment and the power control in two-timescale. For the long-timescale, the vehicle association and resource management variables are optimized. Since the optimization turned out to be NP-hard, to solve it, a MA-VASAPCA was designed. For the short-timescale, Lagrangian dual decomposition was used to determine the power variable that satisfied the instantaneous QoS requirements of various services. Simulation results have shown that the proposed TT-VARMS effectively increases the system sum-rate and decreases the signaling overhead. By adopting the proposed scheme in vehicular networks, the probability of traffic accidents could be significantly reduced while the overall entertainment experiences could be enhanced.

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